SECURE LOCATION TRACKING FOR INTERNET OF THINGS

Capstone Project Proposal

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TABLE OF CONTENTS

- Mentor Consent Form 3
- Project Overview 4
- Problem Statement 4
- Need Analysis 5
- Literature Survey 6
- Objectives 10
- Methodology 11
- Project Outcomes & Individual Roles 12
- Work Plan 13
- Course Subjects 13
- References 14

Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team

Project Title: SECURE LOCATION TRACKING FOR INTERNET OF THINGS		
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Project Overview

The Internet of Things (IoT) has become one of the buzzwords in Information Technology. It is an ecosystem of technologies monitoring the status of physical objects, capturing significant data, and communicating that information to software applications through IP networks. The IoT incorporates multiple long-range, short-range, and personal area wireless networks and technologies into the designs of IoT applications. It integrates many technologies, such as Internet, Zigbee, Bluetooth, infrared, WiFi, GPRS, 3G, etc, which can enable different ways to obtain the location information of various objects.

Location-based service is a primary service of the IOT. Localisation in indoor positioning systems plays an important role in the IoT. Location Based IoT applications range from tracking objects and people in real-time, assets management, agriculture, assisted monitoring technologies for healthcare, and smart homes, to name a few. Many localization algorithms and systems have been developed by means of wireless sensor networks for both indoor and outdoor environments.

The interconnection of location information of these devices brings convenience to people's lives, but there is also the problem of leaking location privacy. Therefore, effective management of the location information of the device is crucial to the development of the Internet of Things. In view of such problems existing in the traditional encryption method in the monitoring of device location information, through this project we design and verify the location information storage system of Internet of things devices based on secure storage based on the research of device location of Internet of things and the key technology of secure storage (like block chain).

Problem Statement

The problem we are addressing is the huge data sensed during localisation on IoT devices which will act as a medium to increase the efficiency of networks in the global world along with secure storage of data which is a major concern because such information can be more susceptible to various attacks. Thus in order to eliminate this, we would like to propose a solution through this project which is to verify whether data coming from sensors is authenticated or not and is stored securely.

Need Analysis

With the rapid development in wireless technologies and the Internet, the Internet of Things (IoT) is envisioned to be an integral part of our daily lives. Localization-based services are among the most attractive applications related to the IoT. They are actually, thanks to the deployment of networks of sensors, able to collect and transmit data in order to determine the targets position. A plethora of localization systems are proposed in the literature. These localization systems are based on different positioning approaches, different techniques and different technologies, making them appropriate for some applications and inappropriate for other applications

The position estimation is an essential component of wireless sensor networks applications [1], especially in the environmental monitoring applications such as [2]: animal dwelling place monitoring, water quality monitoring and farming precision. In addition, the location information enables various emerging applications such as inventory management, intrusion detection, road traffic tracking, health monitoring, etc. [2].

The Internet of Things (IoT) is a paradigm shift in the enhancement of the connectivity of the Internet. It provides the technical feasibility for a huge number of sensors, actuators and smart devices to remain connected with it. It adds several emerging services which are essential for connected living initiatives. Among those value added services, localization is one of the prominent one. Localization has a lot of importance in the connected living environment such as security, surveillance and maintenance of law and order. Localization of moving objects leads to tracking which is essential for similar applications. Both localization and tracking are required for emerging applications in several sectors.

Literature Survey

The Internet of Things conceptualized a big connectivity model where all devices and services from all the places connected to the Internet all time. The 'Things' of IoT are heterogeneous devices consisting of sensors and actuators. The sensor senses signals from the environment and the actuator acts on the environment. The IoT opens an important dimension called 'Location of Things' (LoT) where things get geographical position information. There are billions of 'Things' in IoT and continuously adding. To organize this big data from a huge amount of heterogeneous devices the location information plays an important role. IoT pulls data from huge devices, filters and organizes them by means of location information. Hence location acts as a search engine for organizing big data and devices.

The localization principle developed 20 years back with the development of GPS for military application further it is commercialized. Currently all devices connected to the Internet use location information, e.g., Google Map, Uber, Waze, Foursquare etc. Hence the location based services (LBS) comes into play as a paradigm shift from traditional localization, i.e., GPS. GPS is just a subset of LBS since it cannot localize 'Things' (sensor and actuators) always. There are a huge number of scenarios like indoor network, underwater network, unfavorable weather conditions when GPS does not work. So alternative techniques are required. Hence development of a huge number of techniques and protocols is done for these scenarios and localization plays an important role for services in IoT infrastructure.

A major part of IoT infrastructure consists of heterogeneous sensors so it can be called a sensing service on demand based on the user. IoT infrastructure is a paradigm shift from traditional sensor networks as shown in Fig. 1. It consists of a cloud or virtual sink which schedules and controls a physical sensor network based on the location of users. Traditional WSN has grown into Sensor-Cloud architecture in IoT infrastructure. Here efficient sensor networks are used for Sensing as a Service (SeaaS) model of the IoT infrastructure. The architecture of location based interactive models of IoT infrastructure is presented in Fig. 2. The architecture consists of four layers:

1. Physical WSN Layer: This layer consists of a physical sensor network. The characteristics of these sensors have sensor ID, state, location and ownership.

2. Cloud or Virtual Sink: The sensor layer forwards their sensed data to cloud or virtual sink. The characteristics of cloud are cloud ID, QoS, resource and price. Cloud provides sensing service from different sensor networks from different WSN with different owners. The cloud provides on demand sensing service Sensing as a Service (SeaaS).

3. Application Layer: Application layer provides different applications to the user and access services from the cloud. The characteristics of the application layer is application ID, region of interest, sensor data of interest and Quality of Services (QoS) requirements like sensing intervals, delays etc.

4. User Layer: User layer consists of users which use services through application. The characteristics of users are user ID, location, application ID etc. A single user can use multiple applications.

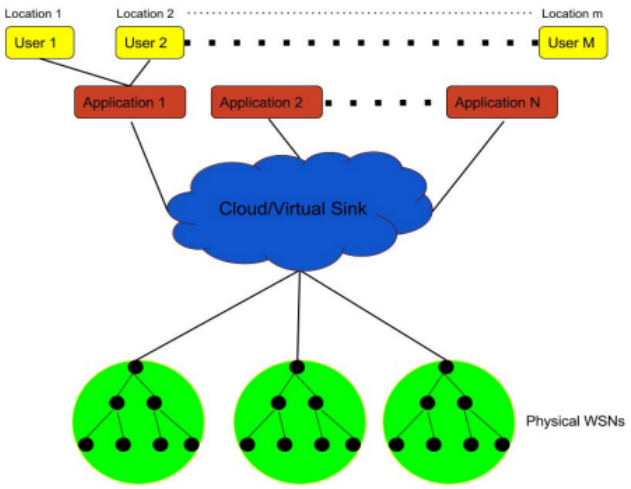


Fig 1. Location based interactive model of IoT (showing Iot-Cloud-WSN integration)

The internet of things strives for creating a world that is fully connected enabling even the tangible objects that are connected with the internet to communicate. It is the incorporation of smart objects of differing types, with the ability to gather information from the physical environment around, compute it, filter it and transmit it over the internetwork [2]. This makes the internet of things to cause dramatic alterations in the everyday life of the people almost in all fields from home, transportation, industries, hospitals, military, entertainment, media, to government conversations etc. [9]. This swift progress of the internet of things has caused collection and transmission of the data in the large scale [4] helping in the tracking, devising, operating, controlling and smart decision making [5] has made it a promising paradigm in a wide range of applications. The data collection of the internet of things is supported by the wireless sensor networks that act as the backbone of the IOT helping in sensing the enormous data from the internet connected tangible objects around. The limitations of the internet of things devices in terms of storage, pushes them for an external storage device to help them in storing the sensed data for the future use. The cloud seems to provide with the improved and the scalable pay as you go storage service for the internet of things as they are equipped with more resources that facilitate computing and communication along with the storage facilities. Despite the promising nature of the cloud storage services for the internet of things, the cloud data centers that assist in the storage of the internet of things suffer from the major issues that is the energy consumption that causes more carbon emission and the security provisioning that maintains the secrecy of the particulars of the persons preventing the illegal access. The researches proceeded with the aim of providing framework that is efficient in the conserving the energy results with the demerit of improper resource usage along with the security issues. The researches with the concept of privacy provisioning for the data located in the cloud causes enormous energy consumption that results with the reduce in the longevity of the network [3] So the placing or locating the data from IOT to the data centers of the cloud requires, procedures that are proficient in terms of energy enhancing the network longevity and secured preventing illegal access against the sensed data that stored in the data centers of the cloud. The paper proposes a multi-objective optimization based on the usage of the resource, energy meeting all the privacy

constraints of the data centers in the location of the sensed data into cloud utilizing the non-dominated sorting genetic algorithm –II and validates the same with the network simulator-II to detail the efficiency and the QOS of the proposed method.

IoT has several features which are suitable for localization and tracking. IoT is a disruptive technology and it is very much ubiquitous. It is essential for sensing and communication at any remote location. In fact, localization and tracking may not be the primary purpose of the IoT deployment. Rather it can be added to any existing IoT as a value added service. This brings beauty into the system. In the smart city initiative, sensors will be deployed in every part of the city for monitoring and service provisioning which are commonly in demand for the citizens. Similarly, there are several other sectors such as manufacturing, healthcare, agriculture, policing, public emergency service provisioning too use localization and tracking. These applications use the locations of the IoT sensors and use them as references to determine the exact locations.

IoT enables the smart devices to collect and exchange data through the network coverage and connections. Localization and tracking have become an integral part of the modern communication systems. Traditional location based services determine the specific location of a movable or unmovable object with the help of GPS through a wireless communication / Internet enabled device. In IoT based localization, it is simplified further and uses the IoT infrastructure to improve the accuracy. Tracking implies determination of the trajectory of an object/animal/human in an environment where network or IoT coverage is available. Localization and tracking are the cornerstones of many applications including efficient security monitoring, green detection, surveillance, smart policing and pet/kids monitoring. These technologies have come into the play due to the need of the time and the advancement in mobile and wireless technologies that provide continuous Internet connectivity. All these together have given an edge to several new business opportunities such as location based devices and services associated with them.

RFID technology allows object identification, localization and real-time tracking without compromising speed or reliability and finds its applications in military purposes. These technologies offer value added services that utilize the navigation tools to locate and track the users. Also the same information can be shared through short message services or any other social platform so that others can also track the users' location. This information is required for common public services such as finding cabs using Google maps. These services are also utilized by the cab service providers such as Uber, Ola and several other taxi companies.

In addition to the advantages, there are several issues in IoT based localization and tracking. Security of users' data is a matter of concern. Security of IoT has not been developed to the required level. Attackers can easily intrude into the IoT networks and disrupt the services. Privacy and security for localization and tracking are still issues to be solved for better reliability and trustworthiness**.**

S.No.	Title	Year	Description
1.	IoT based localization and tracking	2017	In this work, we present the current improvements in localization and tracking using IoT. We start with the differences between the IoT based localization and the non-IoT based systems. Then we show the suitability of IoT for localization and tracking. We show several current applications of IoT in localization and tracking.
2.	Location of Things (LoT): A Review and Taxonomy of Sensors Localization in IoT Infrastructure	2018	In this paper, a recent extensive analysis of localization techniques and hierarchical taxonomy and their applications in different contexts is presented. This taxonomy of the localization technique is classified based on the presence of offline training in localization, namely self-determining and training dependent approaches.
3.	SECURITY AND PRIVACY PRESERVING OF SENSOR DATA LOCALIZATION BASED ON INTERNET OF THINGS	2019	The paper addresses the problem of locating the data sensed from the various applications into the cloud, optimizing the energy consumption of storing the data into the cloud and eluding the illegal access by, proper placement of the conflicting and the non-conflicting in the separate hosts.

Objectives

Localization is utilized in various fields to encourage certain activities e.g security, service provision, navigation and monitoring and tracking. Since the approximate position of objects is the primary thought localization, the objectives talked about give more information about localization.

- Service providing: Service provision in mobile-based traffic free route services and applications that contribute to the service sector called Skilligent [28] since the nodes are constantly moving. Patient care in hospitals, security and maintenance of corporate houses, food services in restaurants and hotels are some of the services provided. Security service provision in homes and industries and tracking services to the users are used in static based because sensors are fixed. Technologies such as GPS, infrared, Internet, WiFi, satellite, Bluetooth all contribute to the process of localization to provide services.

- Navigation: It is mainly mobility dependent. Localization in subsurface territories, submerged or earthbound areas have encouraged navigation in zones which were hard to explore. Navigation in underwater and underground areas using communication among sensor nodes combined with other techniques has made it possible to navigate and collect data. Exchange of information between various systems and vessels [31] likewise in the terrestrial region in which information can be exchanged by the use of IoT equipped vehicles and the main system facilitates easy navigation.

- Security: Major and significant difficulties of IoT is security which may be overcome by some secure storage techniques like storing the information on cloud networks. [29]. Any device equipped with this system should be able to store the information securely.

Methodology

In the last few years research is going on for finding the efficient algorithm for localization. The localization of the sensor node is a two step process of first physical measurement with some reference and second estimate the location from the physical measurement data. Sometimes nodes with limited hardware do not conduct physical measurement rather they try to calculate the position from neighbor node information. The following approaches for localization are existing in the literature. These are based on range measurement, available from anchor nodes, fine-grained, coarse-grained, incremental and concurrent localization methods. The basis of classification of the above scheme is based on either anchor node or range measurement. From the existing literature and IoT application it is found that there is a need for review of existing technology to address the localization problem to the IoT infrastructure. The selection of proper localization methods in the IoT scenario aims to address the restriction and avoid the pitfalls. The proposed taxonomy classifies the localization approach based on offline training. Since IoT infrastructure analyzes huge data and devices. There is a huge scope of application offline training for localization. So the training step is the key for the taxonomic hierarchy of localization methods in IoT infrastructure. The classification is presented in Fig. 3. The localization approach is reviewed into two categories considering the IoT scenario as Self Determining Method Training Dependent Method.

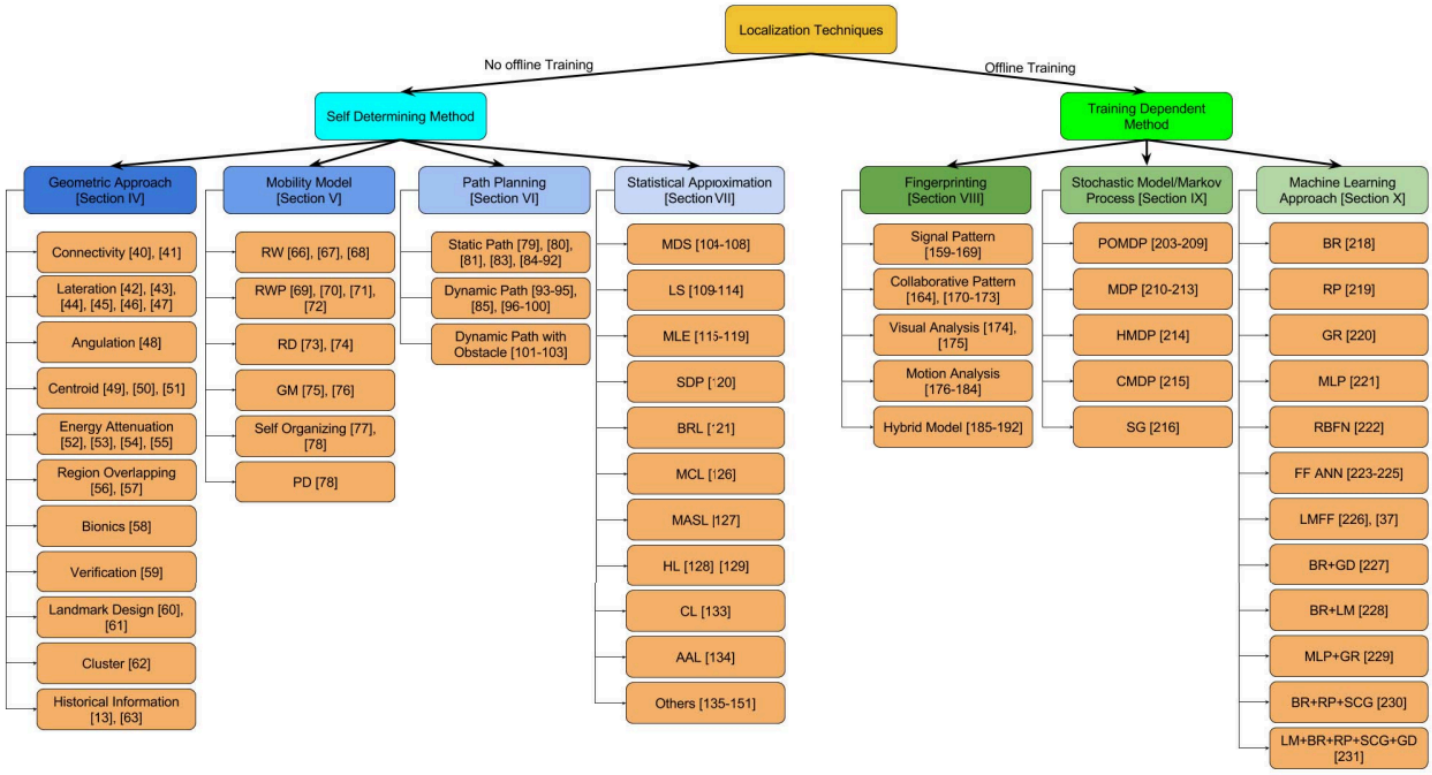


Fig 3. Taxonomy of Localization Methods

Project Outcomes & Individual Roles

The proposed solution offers an embodiment to measure and evaluate the fulfillment of objectives. The embodiments to be offered includes:

1. Real time localization system using IoT.
2. Different methods would be analyzed for the said tasks and most ambient one would be incorporated.
3. The user data would be safeguarded using different security protocols and to provide further security it would be stored at decentralized databases using the blockchain technology.
4. We would try to come up with techniques to scale up the demonstrated ideas.
5. Emphasis would also be laid to incorporate a number of other functionalities to make the system diverse.

Sr. No.	Roll No.	Name	Role/Research Papers
1.	101903311	Manan Ghai	• A survey on mobility-assisted localization techniques in wireless sensor networks. • Skilligent: Skilligent visual localisation system
2.	101903315	Akshat Singh	• Structural health monitoring using wireless sensor networks: a comprehensive survey. • Mobile middleware platform for secure vessel traffic system in IoT service environment. Security and Communication Networks
3.	101903321	Kritarth Gupta	• Locating the nodes: Cooperative localization in wireless sensor networks",IEEE Signal Process. • Mobile middleware platform for secure vessel traffic system in IoT service environment. Security and Communication Networks

Work Plan

Sr. No.	Activity	Month	January		February				March				April				May				June			
		Week No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1.	Identification formulation and Planning of Project	Plan																						
		Actual																						
2.	Study of utility, localization and courses.	Plan																						
		Actual																						
3.	Designing and Development	Plan																						
		Actual																						
4.	Hardware Implementation	Plan																						
		Actual																						
5.	Testing and Performance Analysis	Plan																						
		Actual																						
6.	Results Evaluation	Plan																						
		Actual																						
7.	Final Evaluation	Plan																						
		Actual																						

Course Subjects

- Computer Networks
- Network Programming
- Software Engineering
- Electronic Engineering
- Energy and Environment
- Measurement Science and Techniques

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